Nocturia & Male Incontinence

Condensed rapid revision · facts already answered in Papers 1–4 point to that answer

highlighted number to memorise · **KEY POINT** and **ALGORITHM** boxes carry the reasoning · MCQ exam trivia · † added beyond the source notes · P2 topic = covered in full in that Paper's answer book

Chapter	Topic
Ch 112	Nocturia
Ch 123	Surgical Procedures for Sphincteric Incontinence in the Male
★	Putting it together

112 Nocturia

1 · Definition and terminology

Why this matters: every diary index and trial endpoint depends on what counts as a night void.

- **Nocturia** (International Continence Society, ICS, 2019) is the number of voids during the **main sleep period**, from intending to sleep until intending to rise.
- The sleep period is set by the **individual's own schedule**, not the clock, so shift workers are included.
- The definition does **not exclude 1 void per night** and does **not incorporate bother**.
- ICS 2019 dropped the 2002 phrase "wake **to void**". "Non-urological" causes are therefore **leverage for interdisciplinary care, not grounds to exclude**.
- Guidance comes from the European Association of Urology (EAU) and ICS. The American Urological Association (AUA) has none; its overactive bladder (OAB) guideline calls **isolated nocturia a separate entity**.

Key diary terms

Term	Definition / formula	Meaning
Actual nightly voids (ANV)	Voids in the main sleep period	Severity count
First uninterrupted sleep period (FUSP)	Hours from sleep onset to the first night void	Key sleep measure
Nocturnal urine volume (NUV)	Night voids **plus the first morning void**	Volume made during sleep
Nocturnal polyuria index (NPi)	NUV / 24-h volume	Proportion made at night
Nocturnal urine production (NUP)	NUV / hours of sleep	Hourly rate
Maximum voided volume (MVV); nocturnal MVV (NMVV)	Largest void in 24 h; largest in the sleep period	Proxies for functional capacity
Nocturia index (Ni)	NUV / MVV	>1: night output exceeds capacity
Modified Ni (MNi)	NUV / NMVV	Uses night data only
Predicted nightly voids (PNV)	Ni − 1	Voids expected if the bladder filled fully each time
Nocturnal bladder capacity index (NBCi)	ANV − PNV	>0: patient voids below capacity

Trap: the first morning void is **split**. Its volume goes into NUV, but it counts as daytime frequency, not as nocturia.

2 · Why nocturia matters

Why this matters: nocturia is common, disrupts sleep, causes falls and can be the first sign of systemic disease.

- **It is common**: up to **3 in 5** people >70 years void ≥2 times nightly, and age is the most robust correlate.
- The Boston Area Community Health survey MCQ found higher rates in Black and Hispanic people, partly explained by socio-economic status.

- **Bother** is reported by most people with ≥**2** voids, but **any nocturia that brings a patient to a urologist is clinically significant**.
- **Sleep**: nocturia is the **commonest cause of adult sleep disruption**. Restorative slow-wave sleep (N3) clusters early in the night.
- Most first voids occur within **2–3 h** of sleep onset, so they break slow-wave sleep and cause daytime fatigue.
- **Falls and hip fracture** are more strongly linked to nocturia than to most other lower urinary tract symptoms (LUTS).
- **Systemic disease**: nocturia may herald cardiometabolic (strongest link), renal, endocrine or neurological disease. Two or more voids a night is linked to higher mortality, though causality is unproven.

> **KEY POINT**
> The first void breaks restorative slow-wave sleep; the symptom may be the visible edge of systemic disease.

3 · Pathophysiology — production, capacity and sleep

Why this matters: treatment works only if it targets the mechanism (production, capacity or sleep).

Nocturia is multifactorial; remember **"Sleep CALM"**: **S**leep disorders, **C**omorbidities, **A**ctions (behaviour, lifestyle), **L**ower urinary tract dysfunction (LUTD) and **M**edications. Idiopathic nocturia is a small but real group.

- Every factor upsets **night production**, **functional capacity** or **sleep**. When the bladder reaches capacity the patient either **wakes to void or has enuresis**.
- Sleep disorders cause "convenience" voids in people already awake. The link is **bidirectional**, because voiding also fragments sleep.

Nocturnal polyuria (NP) and global polyuria

Criterion	Threshold	Comment
NPi (ICS)	>**20%** in young adults to >**33%** in older adults	Most used, but discriminates poorly
NUP (ICS alternative)	>**90 mL/h**	More specific for production-driven nocturia
Global polyuria	>**40 mL/kg/24 h** (>2.8 L at 70 kg)	Uncommon but serious; **always investigate**

- **Krimpen study** MCQ: NPi >33% was found in 92% of older men with ≥2 voids, but also in 70% of men **without** nocturia.
- Excess night production contributes to **most adult nocturia**, and the mechanisms often coexist.

Functional capacity and the mismatch

- Normal capacity is usually >300–400 mL. A capacity of **150–300 mL** suggests diminished capacity.
- Diminished capacity may be **24-hour** (all voids are small) or **night-specific** (partly convenience voids).
- **Ni is the strongest diary determinant of severity and of treatment response**.
- PNV equals Ni − 1, because the last night volume leaves the body as the first morning void.

ALGORITHM

Worked example:

1. NUV is 800 mL and MVV is 200 mL, so Ni = 4.
2. PNV = Ni − 1 = 3.
3. The diary shows 6 night voids, so NBCi = 6 − 3 = **3**: the patient voids well below capacity.

Investigate diminished night capacity when NBCi is >**1.3**; a value >**2** is strongly associated with severe nocturia.

4 · The bladder diary

Why this matters: only the diary separates production, capacity and sleep problems.

EAU (strong): use a diary for female LUTS, and for male LUTS with storage symptoms or nocturia.

ALGORITHM

1. Give the patient a measuring container.
2. Record the time and volume of every void.
3. Record the time, volume and type of every drink.
4. Mark bedtime (intending to sleep) and rising time.
5. Grade urgency and note leakage for each void.

- International Consultation on Incontinence (ICI) urgency codes: **0** no need (social void), **1** normal desire, **2** urgency that passed, **3** urgency but dry, **4** urgency with leak.
- **Diary ANV is the gold standard for severity**; questionnaires overestimate it.
- **Intent governs**: a patient lying awake intending to sleep is still in the sleep period.
- **FUSP is the most important sleep measure**; it is both an outcome measure and a treatment target.
- Duration: EAU advises **3 days** (Male guideline: at least 3). Because of "diary fatigue", fewer than half complete it.
- A **nocturnal-only diary** lets MNi substitute for Ni, but it **cannot diagnose global polyuria**. EAU does not recommend it for women.

KEY POINT

Every cause works through production against capacity, gated by sleep; their ratio (Ni) predicts severity better than either alone.

5 · Evaluation — the Sleep CALM work-up

Why this matters: urology is only one of five domains, so a urology-only work-up misses most treatable causes.

Do a basic work-up in all patients, then add tailored tests and multidisciplinary referral. The **TANGO** questionnaire screens all domains, including falls.

Factor	History / examination	Tests (basic, then ancillary)
Sleep	Habits, apnoea, restless legs, pain, anxiety, depression	STOP-BANG and sleep diary; then polysomnography, sleep specialist

Factor	History / examination	Tests (basic, then ancillary)
Comorbidities	Blood pressure, volume overload, neurological signs	Metabolic panel; then HbA1c, ECG, echocardiography, natriuretic peptide, albumin:creatinine ratio, calcium, thyroid, water deprivation test
Actions	Fluid, alcohol, caffeine, sodium; body mass index (BMI)	Fluid intake diary
LUTD	Concurrent LUTS, prior surgery; abdominal, genital, rectal examination	Urinalysis, post-void residual (PVR); then prostate-specific antigen (PSA), uroflowmetry, cystoscopy
Medications	Reconcile drugs and **dose timing**	—

- **Trap**: urology is one row of five. First-line treatment is to **optimise every Sleep CALM abnormality**.

6 · Sleep disorders

Why this matters: treating sleep apnoea gives one of the biggest reductions in night voids of any intervention.

Obstructive sleep apnoea (OSA) is the best-described link; frequent nocturia predicts severe OSA.

Arm	OSA mechanism	Effect of continuous positive airway pressure (CPAP)
Production (main)	Hypoxia causes pulmonary vasoconstriction, which raises right atrial transmural pressure. This releases **atrial natriuretic peptide (ANP)**, causing natriuresis.	Lowers night production
Capacity	Intermittent hypoxia causes oxidative stress and detrusor instability	Less detrusor overactivity
Sleep	More light sleep, less slow-wave sleep	Increases slow-wave sleep

- CPAP reduces nocturia by up to ≈**2 voids** in moderate–severe OSA, which is among the **largest effects of any intervention**. EAU Female (strong): refer suspected OSA to a sleep clinic.
- **STOP-BANG** scores Snoring, Tired, Observed apnoea, Pressure, BMI >**35**, Age >**50**, Neck >**40 cm** and Gender male. A score ≥**3** means high risk; the tool is highly sensitive.
- Ask the **primary motive for getting up**: an urge to void, or already awake. Screen for OSA, restless legs and parasomnias.
- **Melatonin** may suit older adults, and behavioural treatment of insomnia reduced night voids.

> **KEY POINT**
> OSA raises production, lowers capacity and fragments sleep — hence one of the largest nocturia responses to treatment.

7 · Comorbidities

Why this matters: nocturia may be the first sign of heart, kidney or metabolic disease, so it is a reason to look beyond the bladder.

- Comorbidities act mainly through **excess night production**. Nocturia may be the **first sign** of hypertension, metabolic syndrome, coronary disease, diabetic end-organ damage or early kidney disease.
- EAU: consider systemic causes in all patients; even **non-bothersome nocturia merits a frequency–volume chart**.
- **Volume overload** (e.g. heart failure): leg fluid is mobilised on lying down and excreted at night.
- **Hypertension**: blood pressure normally dips **10–20%** in sleep. **Non-dipping**, from renin–angiotensin–aldosterone dysregulation, causes **nocturnal natriuresis**.
- Hydrochlorothiazide helped men whose LUTS were refractory to α-blockers.

Disorders of water homeostasis

Arginine vasopressin (AVP) acts on collecting-duct **V2 receptors**. Defects in this system give **24-hour polyuria**.

Condition	Defect	Causes
Central diabetes insipidus (DI)	Reduced AVP secretion	Brain tumour, haemorrhage, trauma, autoimmune
Nephrogenic DI	Kidney insensitive to AVP	Drugs, electrolytes, renal failure
Primary polydipsia	AVP suppressed by high intake	Idiopathic; psychiatric

Arginine-stimulated **copeptin** appears superior to the water deprivation test. Evaluate every case of global polyuria for neoplastic, traumatic, inflammatory and psychiatric causes.

> **KEY POINT**
> Comorbidity acts through the kidney (third-spaced fluid, nocturnal natriuresis, failed antidiuresis). Global polyuria and non-bothersome nocturia both warrant looking beyond urology.

8 · Actions — lifestyle and behaviour

Why this matters: lifestyle change is cheap, safe and first-line, and when people drink matters more than how much.

- **Timing matters more than amount**. Water causes diuresis within ≈1 h; caffeine and alcohol act within 2–3 h, and both also disturb sleep. Evening protein causes hyperfiltration.
- Restrict fluid, caffeine and alcohol from **3–4 h** before sleep. Older adults should drink more by day. Patients with high baseline intake gain most.
- Other measures are **sodium** restriction, weight loss, exercise, **compression stockings**, bladder training, pelvic floor muscle training (PFMT) and treating constipation.

9 · Lower urinary tract dysfunction

Why this matters: bladder and prostate drugs help nocturia only a little, so set expectations before prescribing or operating.

- Nocturia accompanies bladder outlet obstruction (BOO), OAB (up to **85%** of trial participants), detrusor underactivity, neurogenic bladder and genitourinary syndrome of menopause (GSM) *See Voiding Dysfunction book, Ch 110*.

- **Isolated nocturia is not a LUTS**, so it is not expected to respond to bladder or prostate drugs.
- Do urinalysis in all, and **PVR in both sexes**. Check PSA only if it changes management. †Urodynamics are not routine *See Voiding Dysfunction book, Ch 107*.

Benign prostatic obstruction

- Nocturia is a secondary endpoint of benign prostatic hyperplasia (BPH) trials. **α-blockers** carry most of the effect; 5α-reductase inhibitors (5ARIs) alone are weakest.
- Gains are fractions of a void with a large placebo response, so they are of **statistical, not clinical, significance** *See Voiding Dysfunction book, Ch 113*.
- **Surgery**: transurethral resection of the prostate (TURP) beat microwave therapy, α-blockers and watchful waiting; newer techniques are not proven better. **Counsel** that nocturia may respond sub-optimally.

Overactive bladder

- Night symptoms are the most refractory part of OAB.
- Tolterodine (night voids as primary endpoint) was negative overall but reduced urge-related voids. **Fesoterodine** (nocturnal urgency as primary endpoint) was positive.
- Patients with a **raised NBCi** respond best to an evening dose.
- β3-agonists give a marginal benefit (≈0.1–0.2 voids over placebo).
- **Behavioural therapy** comes first, and it matches or beats drugs.
- Antimuscarinics are on the **Beers Criteria** for people ≥65 years, who are the core nocturia population.

Therapy	EAU Male LUTS	EAU Female LUTS
α1-blocker	Offer for nocturia with LUTS (weak)	—
5ARI	Offer with LUTS and prostate >**40 mL** (weak)	—
Phosphodiesterase-5 (PDE5) inhibitor	**Do not offer**	—
Antimuscarinic	Offer with OAB (weak)	Offer after counselling (**strong**)
PFMT; vaginal oestrogen	—	Offer PFMT (**strong**); offer oestrogen (weak)

KEY POINT
LUT drugs act on capacity and outflow, but most nocturia is production-driven — effects are statistically significant, clinically small. Treat the mechanism, not the organ.

10 · Medications causing nocturia

Why this matters: a drug, or its dose timing, is an easily reversible cause, so reconcile medications in every patient.

Mechanism	Drugs
Increased urine output	Diuretics; selective serotonin reuptake inhibitors (SSRIs, block antidiuretic hormone); **calcium channel blockers** (oedema, raised ANP); tetracycline; lithium (reduces aquaporin-2); **sodium–glucose cotransporter-2 (SGLT-2) inhibitors** (glycosuria)

Mechanism	Drugs
Insomnia / central nervous system effects	Stimulants; α- and β-blockers; salbutamol, theophylline; corticosteroids, thyroxine; SSRIs; phenytoin
Bladder toxicity	Ketamine; tiaprofenic acid; cyclophosphamide

α-blockers both treat nocturia and may cause insomnia. Adjust dose timing first, or substitute another drug.

11 · Management algorithm and refractory nocturia

Why this matters: drugs that cut night urine carry a sodium risk, so they are reserved for refractory cases.

ALGORITHM

1. Take a history, examine, and obtain a diary and basic tests.
2. Identify the Sleep CALM factors.
3. Optimise each factor, with multidisciplinary help.
4. Add conservative measures.
5. If nocturia stays refractory, then consider an antidiuretic or a timed diuretic in selected patients.
6. Repeat the diary at every step.

- **Multicomponent therapy**: treat interacting causes together (EAU **"Shared Care Pathway"**).
- **Target night volume even when capacity is small**: small-capacity patients improved as NUV fell, with capacity unchanged.

Desmopressin

- P3 Desmopressin (DDAVP) — *selective V2 agonist, no pressor effect; hyponatraemia risk factors (age ≥65, Na <135, CrCl <60, high dose), most in first 2–3 weeks; EAU <65 / low dose >65; contraindications*

Not in the paper answer:

- Rationale: AVP normally rises in sleep; some NP patients have a **blunted nocturnal AVP rise**.
- Desmopressin is the **only US-approved nocturia drug**, for **≥2 voids per night due to NP** (level 1b).
- Current ODT doses are sex-specific (table); in men 75 mcg added hyponatraemia, not efficacy.
- †Also contraindicated with systemic or inhaled glucocorticoids and estimated glomerular filtration rate (eGFR) <50 mL/min.
- The US label monitors sodium at baseline, within 7 days and at ≈30 days (EAU schedule below).
- A proposed protocol checks baseline sodium in all, stops if <135, and repeats at 7 and 30 days only if ≥65 or at risk; it would have prevented every case <130 mmol/L at approved doses.
- Future: predominant water diuresis suits desmopressin; predominant natriuresis suits an afternoon diuretic.

	Nasal spray	Orally disintegrating tablet (ODT)
Dose	1.66 mcg if <65 with no risk factors; **0.83 mcg** if ≥65 or at risk	**Sex-specific**: women **25 mcg**; men **50 mcg**
Timing	≈30 min before bed	Sublingual, ≈60 min before bed

- **Women are more sensitive**, because the V2 receptor gene is on the **X chromosome**.

- †Advise minimal fluid from 1 h before until 8 h after the dose.
- Hyponatraemia is defined as sodium < **130 mmol/L**; most cases occur within **3 weeks**.
- **Sodium monitoring** (EAU): baseline, **day 3**, day 7, 1 month, then periodically. EAU Female: avoid if baseline sodium is low.
- A **low dose** still works and avoids antidiuresis outlasting sleep. **Age 65 is the pivot.**

Timed diuretics

- Furosemide acts for 6–8 h. Evening dosing **causes** NP, whereas **morning or early-afternoon** dosing treats it (level 1b).
- Small trials (bumetanide 4–6 h, or furosemide 40 mg 6 h before bed) consistently cut voids. Monitor sodium.
- **Trap**: afternoon furosemide with bedtime desmopressin worked in a trial, but loop diuretics are a **contraindication** to desmopressin in US labelling.

> **KEY POINT**
> Refractory nocturia: move urine production out of the night. The price is sodium balance — dose, age and monitoring define safe use.

12 · Landmark trials

Why this matters: prostate drugs give small nocturia gains; desmopressin dosing is sex-specific.

Trial	Design	Result	Take-home
MTOPS	Doxazosin, finasteride, combination, placebo	Doxazosin and combination beat placebo; finasteride did not	α-blocker drives benefit
VA Cooperative	Terazosin, finasteride, combination, placebo	Only terazosin separated from placebo	5ARI alone does not help
CombAT	Tamsulosin, dutasteride, combination	Combination −0.5 vs −0.3/−0.4 voids	Significant but small
Fesoterodine RCT	Nocturnal urgency primary endpoint	Positive	Helps urgency-driven night voids
Nasal spray phase III	0.83/1.66 mcg vs placebo	Both beat placebo	Basis of approval
ODT sex-specific RCTs	Women 25; men 50/75 mcg	Positive; 75 mcg caused more hyponatraemia	Sex-specific dosing

123 Surgical Procedures for Sphincteric Incontinence in the Male: Artificial Urinary Sphincter and Perineal Slings

1 · Pathophysiology and device mechanisms

Why this matters: how a device works decides which man it suits.

- Male **sphincteric (stress) urinary incontinence (SUI)** needs failure of **both** sphincters. Radical prostatectomy (RP), the commonest cause, removes the internal sphincter, leaving the rhabdosphincter.
- Other causes are radiotherapy (RT), pelvic fracture urethral injury and neuropathy. Treating vesicourethral anastomotic stenosis (VUAS) can damage the membranous urethra.
- Work-up and sling-versus-sphincter overview: *See Voiding Dysfunction book, Ch 106*.

Device	Mechanism	Site	Adjustable	Needs residual sphincter
Artificial urinary sphincter (AUS, AMS 800)	Circumferential compression, cycled by the patient	Bulbar urethra or bladder neck	No	No
Advance / Advance XP	**Repositions** and lengthens the urethra; no compression	Bulbar, transobturator	No	**Yes**
Virtue (quadratic)	Prepubic compression and elevation	Bulbar, four arms	No	Less dependent
ATOMS (adjustable transobturator male system)	Cushion compression	Bulbar, transobturator	Yes, via scrotal port	Less dependent
ProACT	Two para-urethral balloons	Bladder neck	Yes, via scrotal ports	—

- The **AUS** has a cuff (≈2 cm long), a scrotal pump and a pressure-regulating balloon (PRB) that sets cuff pressure.
- The cuff refills passively †over minutes after pumping. Men without detrusor contraction can void by straining or Credé.
- A transobturator sling raises leak point pressure **without obstruction**, by elevating the bulb and rhabdosphincter.

> **KEY POINT**
> Every device compresses the urethra or repositions a functioning sphincter. Trap: ATOMS and Virtue are "slings" but compress — only the Advance depends absolutely on a coapting residual sphincter.

2 · Indications and device selection

Why this matters: severity and radiotherapy history decide between sling and sphincter.

- Indication: **irreversible sphincter incompetence** with bothersome incontinence (UI) after conservative care.
- Do not delay surgery beyond **6 months** for severe SUI, especially when cystoscopy shows a sphincter defect.
- 24-hour pad weight grades severity: mild **<150 g**, moderate **<400 g**. AUS is preferred if >250 g or after RT (MASTER trial *See Ch 106*).

Stratum (American Urological Association and Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction)	Recommendation
Severe UI or **any RT**	AUS first-line; sling only with counselling
Moderate, no RT	AUS or male sling; adjustable balloons in specialised centres
Mild, no RT	Continue pelvic floor muscle training (PFMT), sling, possibly AUS

- Counsel: ~1 thin pad/day, AUS lifespan **7–10 years**; failed slings mostly go to AUS.
- **A sling needs** mild–moderate SUI, a contractile bladder (if hypocontractile, choose AUS), a non-irradiated urethra and a coapting sphincter.
- **An AUS needs** manual dexterity and cognition, a healthy urethra and adequate bladder compliance.

Absolute — need an alternative	Relative
Low compliance threatening the kidneys; low capacity (urgency incontinence); urethra too poor. Alternatives: clamp, catheter, bladder neck closure, diversion	Frequent transurethral work (bladder cancer, refractory VUAS, radiation cystitis), because it risks erosion and infection

- Radiation cystitis needs hyperbaric oxygen and **6 months** free of bleeding before AUS.

> **KEY POINT**
> Each device's contraindication mirrors the other's requirement.

3 · The high-risk ("fragile") urethra

Why this matters: erosion destroys the urethra, so spot the high-risk man first.

- Strong erosion risk factors are **prior RT, prior erosion or infection, and urethroplasty**.
- Others: hypogonadism, catheterisation, 3.5-cm cuff after RT, tandem distal cuff. Debated: coronary disease, age >80, diabetes, smoking.
- After a prior erosion, 5-year repeat erosion approaches **50%**.
- Urethroplasty is worse than RT, because transection divides the bulbourethral arteries.
- Countermeasures: optimise medically, use a **transcorporal (TC) cuff**, a PRB of **51–60 cm H_2O**, a cuff 0.5 cm larger, and nightly deactivation.

> **KEY POINT**
> Erosion = spongiosal ischaemia under constant pressure. Countermeasures add tissue (TC tunica) or lower pressure — never squeeze harder.

4 · Operative technique

Why this matters: cuff site, size and pressure decide continence and erosion.

- Low lithotomy, chlorhexidine–alcohol, sterile urine. Give IV antibiotics within **60 min** of incision: aminoglycoside 5 mg/kg plus a first/second-generation cephalosporin or vancomycin.
- Place the bulbar cuff on the **proximal bulbar urethra, proximal to corporal fusion**. Spongiosum is thickest here, sitting does not open the cuff, and distal sites stay free for revision.

ALGORITHM

1. Midline perineal incision; divide bulbospongiosus.
2. Incise Buck fascia at the crural divergence and make a 2-cm tunnel around the urethra.
3. Measure and place the cuff (usually 4.0–4.5 cm).
4. Place the PRB preperitoneally (lower-quadrant incision or via the external ring) and fill it with **23 mL**.
5. Place the pump subdartos in the anterior scrotum.
6. Cycle the device; do urethroscopy.
7. Leave it **deactivated for 4–6 weeks**.

- Sizing: use a consistent method; less tight cuffs continent better. A **3.5-cm cuff** raises erosion after RT.
- Avoid the blind external-ring PRB route after mesh hernia repair or cystectomy.
- †Deactivate (open) the cuff before any later catheterisation or endoscopy.

PRB (cm H_2O)	Use
51–60	High-risk bulbar (RT, prior erosion, atrophy); trades continence for safety
61–70	Standard bulbar cuff
71–80	Bladder neck cuff

- **Tandem cuff**: a second cuff ≥1.5–2 cm distal, joined by a Y-connector, deactivated 6 weeks. It is not first-line, because the distal cuff erodes more.
- **Transcorporal cuff** (Webster 2002 MCQ): the dissection goes through the corpora, so ventral tunica stays on the dorsal urethra and adds bulk.
- Use it for a urethra <3.5 cm, revisions, prior erosion or urethroplasty. Erections are usually preserved; transient retention is commoner; erosion benefit is unproven.
- **Bladder neck AUS** suits an intact prostate (neuropathic bladder, myelomeningocele) and is **contraindicated after RP**. Cuff ≥8 cm, PRB 71–80; erosion is less.

ALGORITHM

Advance XP:

1. Perineal incision; open bulbospongiosus.
2. Divide the perineal body to release the bulb.
3. Pass the helical trocar outside-in below the adductor longus tendon.
4. Fix the mesh to the spongiosum.
5. Tension it under cystoscopy until the mucosa coapts (the bulb moves 3–4 cm proximally).

- **ATOMS**: bulbospongiosus is **left intact**; the scrotal port allows adjustment in clinic. **ProACT**: balloons are placed at the bladder neck under fluoroscopy.

5 · AUS complications and management

Why this matters: each complication must leave a usable urethra.

ALGORITHM

Retention:

1. Confirm the cuff is deactivated.
2. Insert a small catheter for 24–48 h.
3. If the second trial fails, then place a **suprapubic cystostomy**, avoiding the PRB.
4. If retention lasts weeks, then the cuff is undersized: replace it.

- **Infection** occurs in **1–3%** (up to 10% if high-risk), usually staphylococci. **Scrotal pain is the key early sign**.
- If antibiotics fail, explant all and reimplant at 3–6 months. Immediate salvage (remove, irrigate, new device) is only for men not eroded, septic or immunosuppressed. Coated devices have not reduced infection.
- **Erosion** occurs in **2–15%** (up to 35% with several risk factors). Remove **all components**.
- If <50% of the urethral circumference is lost, leave a catheter ~3 weeks; if >50%, do in situ ventral approximation or anastomosis. Stricture follows erosion in 8–35%.
- Reimplant after **3–6 months** at a **new site**, using a TC cuff with or without a low-pressure PRB.
- **Atrophy** causes a **gradual** recurrent leak and is a common reason for revision. Downsize, relocate (TC cuff distally) or add a tandem cuff. **Never simply raise PRB pressure**.
- **Mechanical failure**: within 3 years, replace the failed part; **after 3 years, replace the entire device**.

Recurrent leak after AUS	Clue, then how to confirm
Deactivated	Sudden; reactivate
Fluid leak	Sudden; PRB shrinks on imaging
Atrophy	Gradual; PRB volume normal (>20 mL)
Erosion / detrusor overactivity	Cystoscopy / urodynamics

KEY POINT

Sudden = device, gradual = urethra. Manage every complication to keep a salvageable urethra for the next cuff.

6 · Sling complications and failure

Why this matters: the AUS rescues failed slings.

Device	Commonest	Erosion / explant
Transobturator	Transient groin pain; retention or urgency 4–10%	0–0.4%
ATOMS	More complications	Erosion/infection ~10%; explant 10–16%
ProACT	**Perforation** (~5%); migration; revision 22%	~4%

- Cystoscope every worsening sling. After a failed sling the **AUS is the gold standard**, with results as good as primary AUS. Repeat Advance only if AUS is refused (~56%).

7 · Special circumstances

Why this matters: strictures, penile prostheses and climacturia change the plan.

- A stricture at the cuff may herald erosion. Treat VUAS above the cuff by balloon or laser via a flexible scope.
- AUS with a penile prosthesis: synchronous and metachronous placement do equally well; if staged, place the AUS first.
- **Climacturia** is treated with a sling or AUS; with erectile dysfunction (ED), add a **mini-jupette** graft.

8 · Outcomes

Why this matters: cure is similar across devices.

- P2 P3 Post-prostatectomy incontinence; SUI: evaluation in females, role of UDS, and surgical therapy in males — *AUS social continence 55–77%, satisfaction ~90%, revision 10–29% at 3 years; Advance cure ~60%; ATOMS ~two-thirds dry*

Not in the paper answer:

- ProACT is ~60–67% dry; cure clusters at 55–65% for all devices.
- Advance XP cure is durable but poorer after RT, in severe UI and after prior surgery.
- Devices differ in durability, erosion and eligibility, not in cure.

★ Putting it together

A · Nocturia: three diary questions

Ask these in order of danger.

ALGORITHM

1. **Global polyuria?** If >40 mL/kg/24 h, then think of diabetes, diabetes insipidus or polydipsia; always investigate.
2. **Nocturnal polyuria?** If the night fraction is >20% (young) or >33% (older), then look for heart failure, sleep apnoea, oedema, evening fluids; consider desmopressin with sodium checks.
3. **Reduced night capacity?** If NBCi >1.3, then treat overactive bladder, obstruction or residual. Screen sleep in all ("Sleep CALM") (*See Ch 112*).

B · Post-prostatectomy stress incontinence

Severity and radiotherapy decide sling or sphincter.

KEY POINT

Pelvic floor training comes first; severe leakage needs surgery by ~6 months. 24-h pad weight: mild <150 g, moderate <400 g. Mild–moderate leakage with a contractile detrusor and no radiotherapy suits a sling. Leakage >250 g, radiotherapy or a failed sling needs an artificial sphincter. Erosion is spongiosal ischaemia (radiotherapy, prior erosion, urethroplasty): add tissue (transcorporal cuff) or lower pressure, never tighten. † Remove eroded or infected devices (*See Ch 123; Ch 106*).